

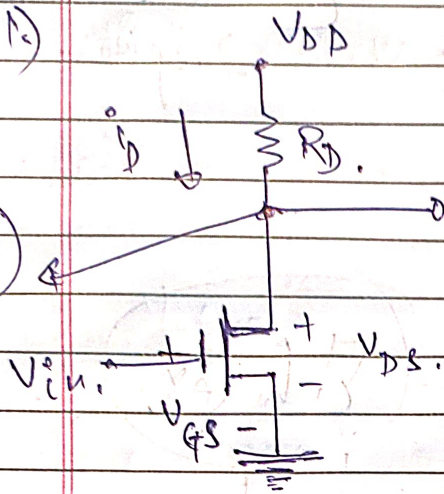
Romit Dutta,

1010683

Date 10/10/2025.

saathi

NANO ELECTRONICS (H/w)



$$V_{DSsat} = V_{GS} - V_t$$

① $V_{DS} > V_{DSsat}$ [saturation zone]

② $V_{DS} = V_{DSsat}$ [onset of pinchoff]

③ $V_{DS} < V_{DSsat}$ [ohmic/linear zone]

(1R + 1 NMOS circuit)

we know, $V_{DD} = I_D R_D + V_{DS} + 0$
(drop 1) (drop 2) (ground).

$$I_D = \frac{V_{DD} - V_{DS}}{R_D}$$

$$R_D = 5K\Omega$$

$$V_{DD} = 2.5V$$

$$R_{D2} = 25K\Omega$$

$$I_D = \frac{2.5 - V_{DS}}{5 \times 10^3}$$

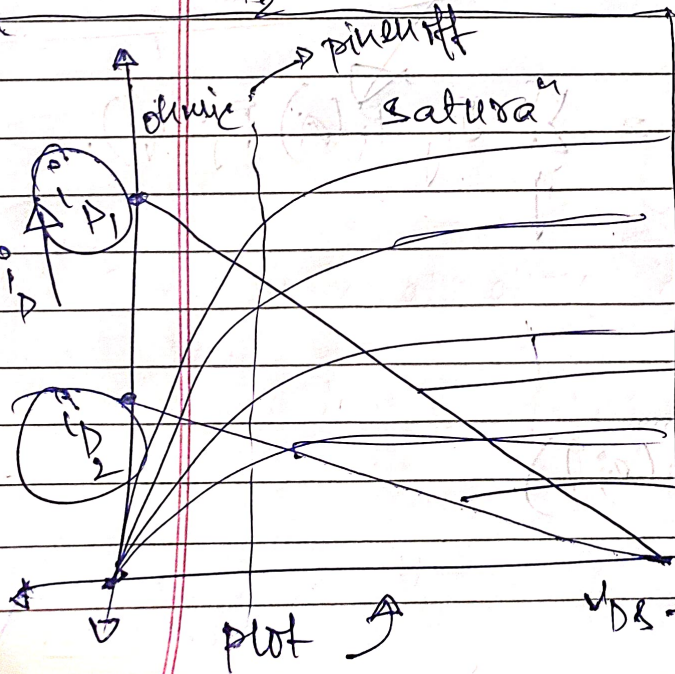
$$I_{D1} = 0.5 \times 10^{-3} - 0.2 \times 10^{-3} V_{DS} \quad (i)$$

$$I_{D2} = \frac{2.5 - V_{DS}}{25 \times 10^3}$$

$$I_{D2} = 0.1 \times 10^{-3} - 0.4 \times 10^{-4} V_{DS} \quad (ii)$$

$$S_1 = (-0.2 \times 10^{-3})$$

$$S_2 = (-0.4 \times 10^{-4})$$



Family of curves [different V_{GS}]

(b) $V_{out} = V_{DS}$, $V_{in} = V_G$

Transfer characteristic

Now $I_{GS} \rightarrow$ (1) (2) $V_{GS} < V_t$ [cutoff].

$\therefore \beta = \mu C_{ox} W/L$

(2) $\beta (V_{GS} - V_t - \frac{V_{DS}}{2}) V_{DS}$

$= \beta \left[(V_{out} - V_t) V_{DS} - \frac{1}{2} V_{in}^2 \right]$ (linear)

$\therefore \frac{V_{DD} - V_{DS}}{R_D} =$

(3) $\frac{\beta}{2} (V_{in} - V_t)^2$ (saturation)

For linear region.

$\therefore \frac{V_{DD} - V_{DS}}{R_D} = \beta \left[(V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$

$V_{DD} - y = (\beta R_D) \left[(x - V_t) y - \frac{1}{2} y^2 \right]$ $\left\{ \begin{array}{l} y = V_{DS} \\ x = V_{GS} \end{array} \right.$

$\Rightarrow \frac{V_{DD}}{\beta R_D} - \frac{y}{\beta R_D} = xy - V_t y - \frac{1}{2} y^2$

$\frac{y^2}{2} + y \left(V_t - \frac{1}{\beta R_D} - x \right) + \frac{V_{DD}}{\beta R_D} = 0$

$\therefore y = - \left[V_t - \frac{1}{\beta R_D} - x \right] \pm \sqrt{\left(V_t - \frac{1}{\beta R_D} - x \right)^2 - \frac{2V_{DD}}{\beta R_D}}$

$V_{out} = - \left[V_t - \frac{1}{\beta R_D} - V_{in} \right] \pm \sqrt{\left(V_t - \frac{1}{\beta R_D} - V_{in} \right)^2 - \frac{2V_{DD}}{\beta R_D}}$

Date ____ / ____ / ____

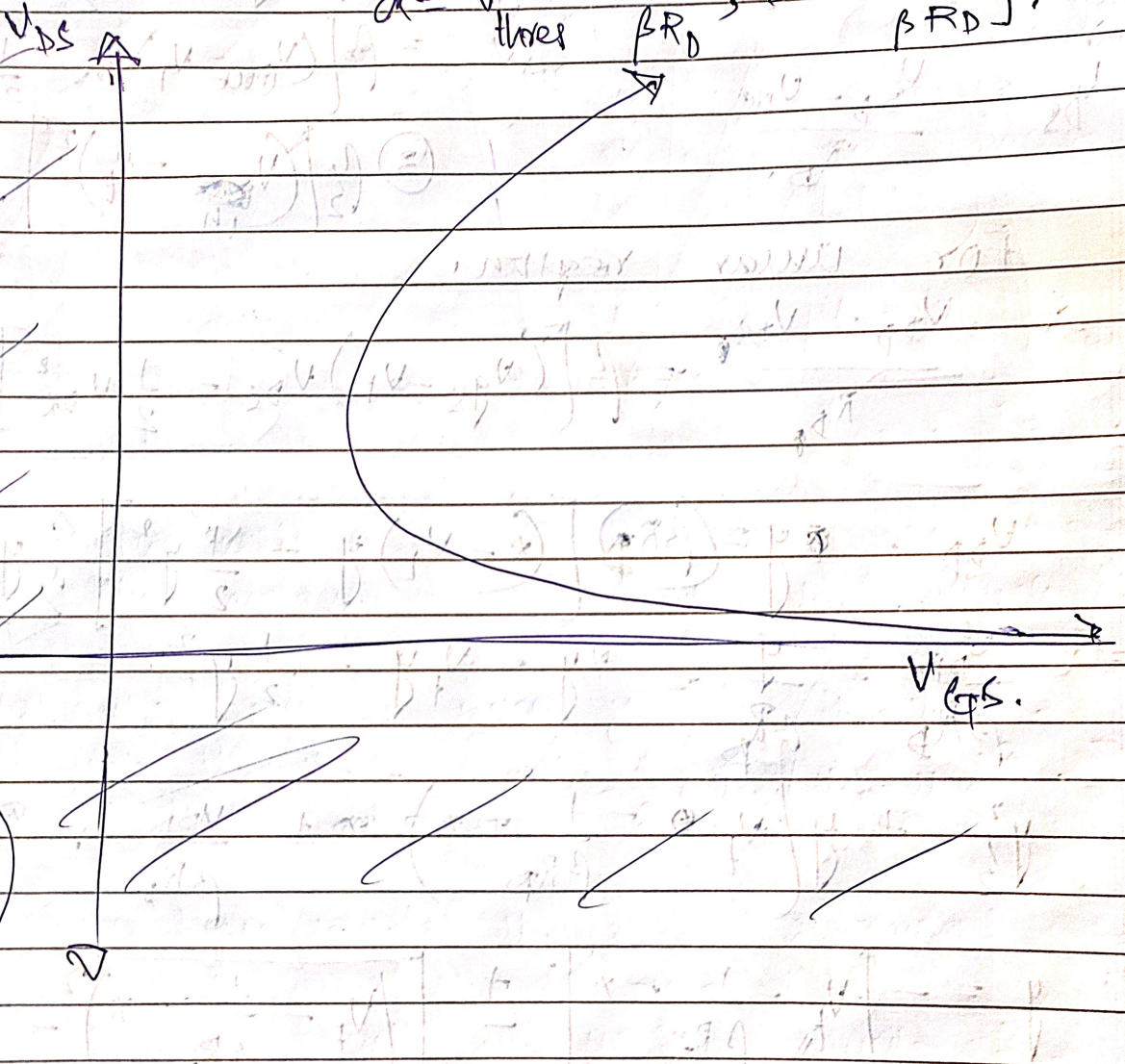
In the linear zone, the plot is hyperbola.

of the form: $y^2 + y(a-x) + b = 0$.

(b > 0)

$$\left[y = V_{DS}/V_{DD}, x = V_{GS}/V_{th} \right]$$

$$a = V_{th} - \frac{1}{\beta R_D}, b = \frac{V_{DD}}{\beta R_D}$$



In the saturation zone,

$$\frac{V_{DD} - V_{out}}{R_D} = \beta \left[(V_{in} - V_t)^2 \right]$$

$$V_{out} = V_{DD} - \frac{1}{\beta R_D} \left[(V_{in} - V_t)^2 \right]$$

This is a quadratic plot of form,

$$y = a - b(x - c)^2$$

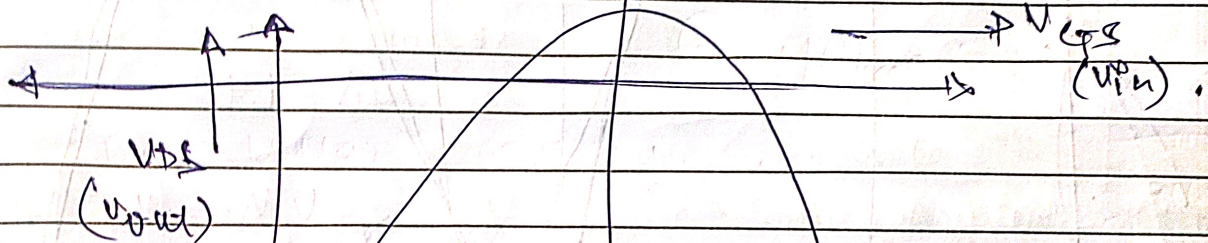
$$\left[\begin{array}{l} a = V_{DD} \\ b = \beta R_D \\ c = V_T \end{array} \right]$$

$$a > 0$$

$$b > 0$$

$$c > 0$$

Plot variation :-

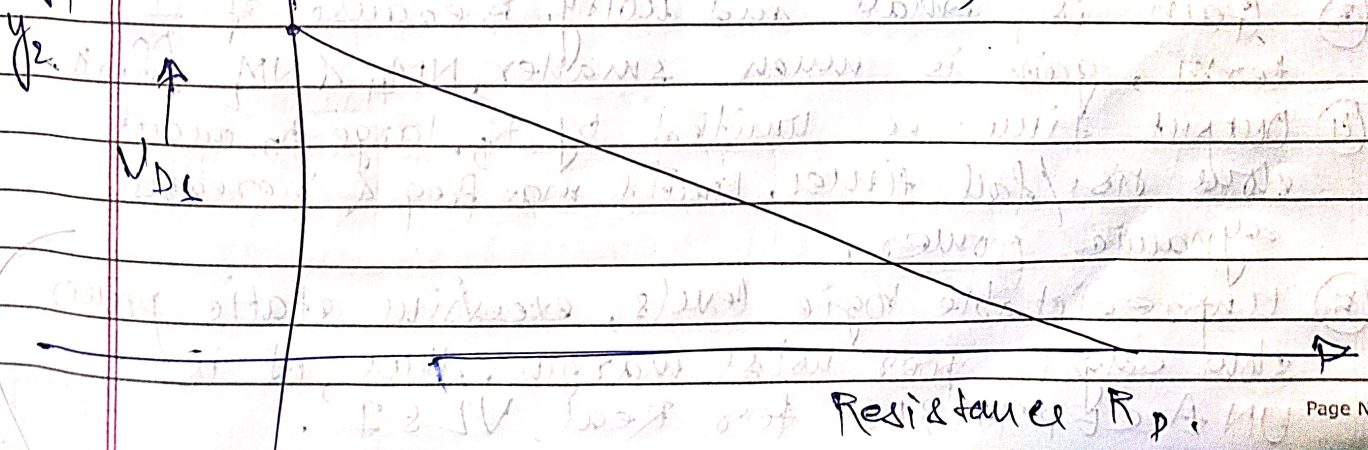


for low resistance, the width is big.
for high resistance, the width is small.

In ~~linear~~ saturation region if V_{GS} is constant.

$$y = a - cx \quad \left[\begin{array}{l} y = V_{DS}, x = \beta (V_{in} - V_T)^2 \\ a = V_{DD} \end{array} \right]$$

$$y_1 = a - cx_1$$

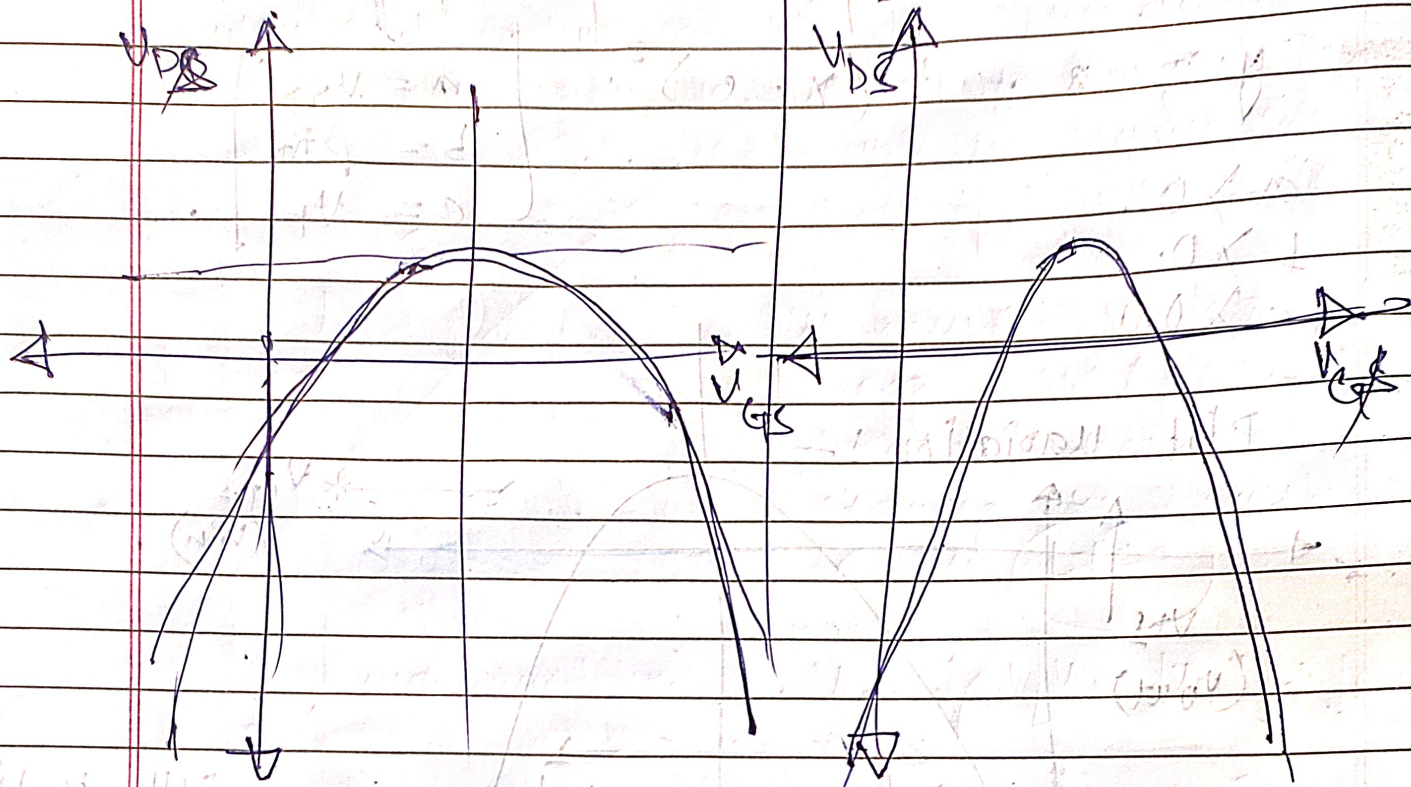


Resistance R_D

Date ___ / ___ / ___

$$R_1 = 5k\Omega$$

$$R_2 = 25k\Omega$$



c) ① When input is on, DC current flows through R_D . The current $i_D = \frac{1}{R_D} (V_{DD} - V_{out})$, which in

saturation zone is given by $\frac{1}{2} \beta (V_{in} - V_{th})^2$. So with $V_{th} = V_{DD}$, Power = $\frac{1}{2} \beta V_{DD} (V_{DD} - V_{th})^2$. This is NOT negligible.

② Poor logic levels. V_{OL} is NOT guaranteed to be close to 0, and depends on R_D & β .

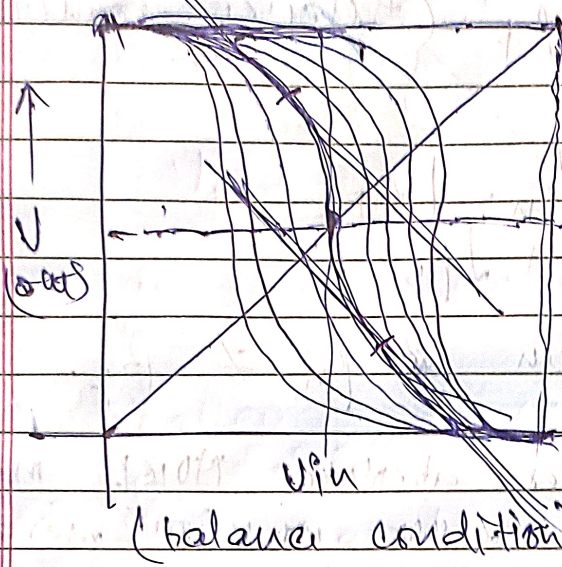
③ Gain is small and noisy. Because of $\frac{1}{R_D}$ term, gain is much smaller, NM_H & NM_L shrink.

④ Output time is limited by R_D . large R_D means slow rise/fall times, limits max freq & increases dynamic power.

⑤ Unpredictable logic levels, excessive static power, slow edges, poor noise margin. Thus, it is UNACCEPTABLE for Real VLSI.

Date ___ / ___ / ___

2(a) skewing in a CMOS inverter refers to the intentional adjustment of the switching voltage. This can be done by varying the strength coefficient ratio of PMOS (β_p) & NMOS (β_n).



We consider the case of strong NMOS ($\beta_n > \beta_p$). Here V_M decreases, showing the transfer plot to the LEFT.

$$V_M = \frac{\sqrt{\beta_p} [V_{DD} - |V_{tp}|] + \sqrt{\beta_n} |V_{tn}|}{\sqrt{\beta_p} + \sqrt{\beta_n}}$$

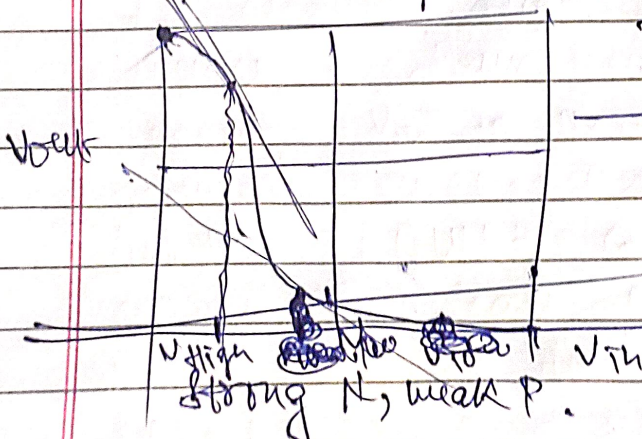
(b) skewing alters NOISE margins.

- Noise margin for logic High: $NM_H = V_{OH} - V_{IH}$.
- Noise margin for logic Low: $NM_L = V_{IL} - V_{OL}$.

V_{OH} & $V_{OL} \sim V_{DD}$ & 0. (ideally).

V_{IH} & $V_{IL} \sim$ input voltages where $\frac{dV_{out}}{dV_{in}} = -1$.

In the operating scenario of ($\beta_n > \beta_p$).



V_{IL} decreases & V_{IH} decreases.

$$NM_H = V_{OH} - V_{IH}$$

Since the graph is skewed to the left, both V_{IL} & V_{IH} decreases.

Date ____ / ____ / ____

Now $NM_H = V_{OH} - V_{IH}$ (V_{IH} decreases, NM_H increases)
 $NM_L = V_{IL} - V_{OL}$ (V_{IL} decreases, NM_L decreases)
 Hence, for logic NM increases if the other decreases

(3)

- (a) The short circuit current I_{sc} flows when both NMOS & PMOS are simultaneously in the saturation region, during input switching. This occurs when the input voltage V_{in} is BETWEEN V_{TH} and $V_{DD} - |V_{tp}|$. This creates a direct path from V_{DD} to ground.
 For slow rise/fall times (t_r & t_f), the input spends more time in the voltage region, prolonging the period both transistors are on. The short-circuit current per switching event is approximated as:

$$I_{sc} = \frac{\beta_1}{12} (V_{DD} - 2V_{TH})^2 \cdot \frac{t_r}{T}$$

Since, $I_{sc} \propto t_r$, slower switching increases I_{sc}

- (b) Dynamic power consumption arises from charging & discharging the load capacitance (C_L)

Power dissipated / switching cycle $\propto C_L V_{DD}^2 f$

$\therefore P \propto C_L$, hence larger C_L increases power consumption

Date ___/___/___

- (4.) Constant Electric field (CE) scaling, based on DENNARD'S SCALING RULES, scales all dimensions (L , W , oxide thickness) by a factor ≤ 1 , voltages (V_{DD} , V_{th}) by ≤ 1 while doping concentrations increase by ≥ 1 .

$$V_t = V_{FB} + 2\phi_F + \frac{2qN_A t_{ox}}{C_{ox}} (2\phi_F)$$

Thus, threshold voltage also ^{can} increase.

5. Challenges of CMOS scaling:

1. Short channel effect.
2. Threshold voltage variation.
3. DIBL (Drain induced barrier lowering).
4. Gate leakage current.
5. GIDL (Gate induced ~~drain leakage~~ ^{drain leakage} ~~current~~).
6. ~~Channel S/D~~ - parasitic resistance.
7. Mobility issues / velocity saturation / Hot carrier effect.

Broadly the problems associated exclusively from scaling are $\rightarrow$

1. Short channel effect
As channel length $\propto 100\text{ nm}$, depletion region of source & drain penetrate into the channel causing V_{th} to decrease. High Drain voltage reduces source-channel potential barrier [DIBL], further LOWERING V_{th} & increasing leakage current. This leads to subthreshold leakage, even at $V_{in} = 0$, causing higher static power consumption.

(MITIGATION) Multigate devices like FINFETs & GAAFETs

improve control. Silicon-on-insulator technology (SOI) can isolate transistors & reduce leakage.

(2) Gate oxide tunneling & GIDL:

As devices scale, gate oxide thickness also decrease. However when SiO_2 thickness falls below 1nm, electrons can tunnel through the oxide layer, creating significant gate leakage current. This contributed to static power consumption and can damage the oxide.

MITIGATION Replace SiO_2 with high- κ dielectric materials (HfO_2 , LaO_2) that provide equivalent oxide thickness (EOT) with greater thickness. Implement metal gates (TiN, TaN) to eliminate polysilicon depletion effects.

(3) HOT carrier injection:

In short channel devices at high V_{DS} (applied) carriers gain high kinetic energy near drain. These "hot carriers" can tunnel into oxide layer & be stuck there. Such carriers cause V_{th} shift & mobility issues. ~~High electric field (vertical), mobility degrades.~~

MITIGATION LDD (lightly doped drain) to reduce peak electric fields near the drain. Graded junctions to spread out high field region.

→ Prolonged exposure to high electric fields (vertical) and high doping cause "mobility (carrier) degradation". Additionally, at high lateral fields carrier velocity saturates.

(4) ~~Penetration~~ Punch through effect.

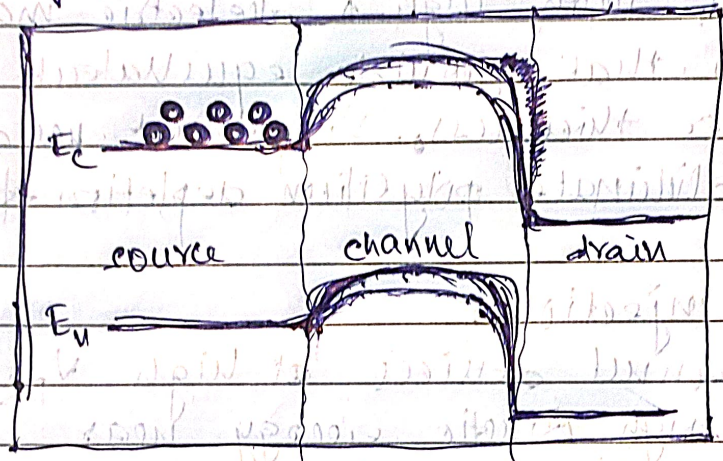
In extremely short channel devices, source & drain depletion regions can merge creating a continuous depletion path that bypasses gate control entirely.

Date ____/____/____

(Poor) When gate control is lost, drain current becomes independent of gate voltage. This causes extremely high leakage and complete device failure.

MITIGATION: ~~increasing~~ increasing doping in channel causes smaller / narrower depletion regions.

(6) Long-channel MOSFET



(*) Short-channel MOSFET

